

Application of PDE

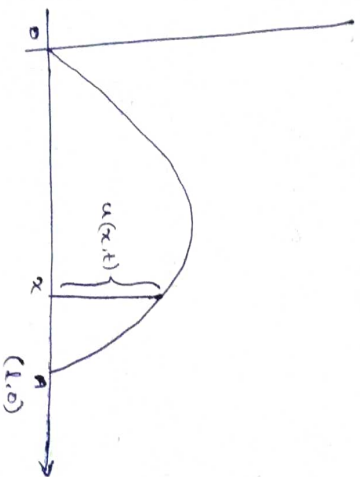
Module - IV

1. 1-D wave equation.
2. 1-D Heat equation.
3. Laplace equation.

→ Method of separation of variables,
→ D'Alembert's solution of wave equation.

1. Formulate as a PDE with bc & ic. (bc $\Rightarrow$ boundary condition)
2. Fourier series solution.
3. Derive the constants using given condition.

1-D wave equation.



Consider an elastic string;

At end points, displacement = 0.

$$u(0, t) = u(1, t) = 0. \quad (bc)$$

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

$$ic. \begin{cases} u(x, 0) = f(x), \\ \frac{\partial u}{\partial t} \Big|_{t=0} = g(x). \end{cases}$$

Consider an elastic string stretched along the x -axis of length l placed along the x -axis whose ends are fixed at $x=0$ and $x=l$. Let $f(x)$ and $g(x)$ respectively be the initial displacement & initial velocity of the string. If $u(x,t)$ is the displacement of the string at any point x & at any instant $t > 0$ then, the complete initial boundary value problem is

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} ; 0 < x < l, t > 0$$

Boundary conditions; — (2)

$$u(0,t) = 0, \quad u(l,t) = 0, \quad t \geq 0 \quad \text{--- (2)}$$

and initial conditions; — (3)

$$\left. \begin{aligned} u(x,0) &= f(x), \\ \frac{\partial u}{\partial t} \bigg|_{t=0} &= g(x), \quad 0 \leq x \leq l. \end{aligned} \right\} \text{--- (3)}$$

The general solution of 1-D wave equation with boundary conditions (2) & initial conditions (3) is given by

$$u(x,t) = \sum_{n=1}^{\infty} \left[C_n \cos \frac{n\pi ct}{l} + D_n \sin \frac{n\pi ct}{l} \right] \cdot \sin \frac{n\pi x}{l}$$

$$\text{where, } C_n = \frac{2}{l} \int_0^l f(x) \sin \left(\frac{n\pi x}{l} \right) dx, \quad n = 1, 2, 3, \dots$$

$$D_n = \frac{2}{n\pi c} \int_0^l g(x) \sin \left(\frac{n\pi x}{l} \right) dx, \quad n = 1, 2, 3, \dots$$

2) If the string is released from rest then,

$$\frac{\partial u}{\partial t} \bigg|_{t=0} = g(x) = 0.$$

then $D_n = 0$ for all.

$$\therefore u(x,t) = \sum_{n=1}^{\infty} C_n \cos \frac{n\pi ct}{l} \cdot \sin \left(\frac{n\pi x}{l} \right)$$

b) If the string is initially in an equilibrium position,

$$u(x,0) = f(x) = 0.$$

$$\therefore C_n = 0 \text{ for all.}$$

$$\therefore u(x,t) = \sum_{n=1}^{\infty} D_n \sin \left(\frac{n\pi ct}{l} \right) \cdot \sin \left(\frac{n\pi x}{l} \right).$$

1. A finely fixed stretched string with fixed end points $x=0$ & $x=l$ is initially in its equilibrium position. If it is set vibrating by ~~twisting~~ ^{giving} a velocity $v_0 x(l-x)$ at any time t .

Ans. Let $u(x,t)$ be the displacement fnc of string then, 1-D wave eqn is)

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

$$\text{with BC } \Rightarrow u(0,t) = 0, \quad u(l,t) = 0$$

$$\& \text{ IC } \Rightarrow u(x,0) = f(x), \quad \frac{\partial u}{\partial t} \bigg|_{t=0} = g(x).$$

$$\therefore u(x,0) = 0 = f(x) \quad (\text{and case}).$$

$$\therefore \frac{\partial u}{\partial t} \bigg|_{t=0} = v_0 x(l-x).$$

$\therefore$ solution is given by;

$$u(x,t) = \sum_{n=1}^{\infty} D_n \sin \left(\frac{n\pi ct}{l} \right) \sin \left(\frac{n\pi x}{l} \right)$$

$$\text{where, } D_n = \frac{2}{n\pi c} \int_0^l g(x) \sin \left(\frac{n\pi x}{l} \right) dx.$$

$$g(x) = u_0 x (l-x) \quad , \quad n = 1, 2, 3, \dots$$

$$\therefore D_n = \frac{2}{n\pi c} \int_0^l u_0 x (l-x) \sin\left(\frac{n\pi x}{l}\right) dx.$$

$$= \frac{2u_0}{n\pi c} \int_0^l (lx - x^2) \sin\left(\frac{n\pi x}{l}\right) dx.$$

$$= \frac{2u_0}{n\pi c} \left[(lx - x^2) \cdot \frac{-\cos \frac{n\pi x}{l}}{\frac{n\pi}{l}} - \frac{2lx - x^3}{3} \cdot \frac{-\sin \frac{n\pi x}{l}}{\frac{n\pi}{l}} \right]_0^l$$

$$= \frac{2u_0}{n\pi c} \left[0 - \frac{l^3}{2} \cdot \frac{l^3}{3} \right]$$

$$= \frac{2u_0}{n\pi c} \left[-\left(lx - x^2\right) \frac{l}{n\pi} \cos \frac{n\pi x}{l} + \left(\frac{l^3}{n\pi}\right) (l-2x) \sin\left(\frac{n\pi x}{l}\right) - \frac{2}{n\pi} \cos\left(\frac{n\pi x}{l}\right) \right]_0^l$$

$$= \frac{2u_0}{n\pi c} \left[2\left(\frac{l}{n\pi}\right)^3 \cos n\pi - 1 \right]$$

$$= \frac{4u_0}{n^4\pi^4 c} \left[(-1)^n - 1 \right].$$

when n is even, $D_n = 0$.

$$\therefore u(x, t) = \sum_{n=1}^{\infty} \frac{4u_0 l^3}{n^4\pi^4 c} \left[(1 - (-1)^n) \sin\left(\frac{n\pi x}{l}\right) \sin\left(\frac{n\pi t}{l}\right) \right]$$

2. A tightly stretched flexible string has its ends fixed at

$x=0$ & $x=l$. At time $t=0$, the string is given a shape defined by $f(x) = u_0 x (l-x)$ where u_0 is a constant and then released. Find displacement at any point x of the string at any time $t > 0$.

Ans. $\frac{\partial^2 u}{\partial t^2} = c^2 \cdot \frac{\partial^2 u}{\partial x^2}$, $0 \leq x \leq l$, $t > 0$.

with BC: $u(0, t) = u(l, t) = 0$, $t > 0$.

IC: $u(x, 0) = f(x) = u_0 x (l-x)$.

$$\left. \frac{\partial u}{\partial t} \right|_{t=0} = g(x) = 0.$$

$$\therefore u(x, t) = \sum_{n=1}^{\infty} c_n \cdot \cos\left(\frac{n\pi t}{l}\right) \sin\left(\frac{n\pi x}{l}\right)$$

where, $c_n = \frac{2}{l} \int_0^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx$.

$$= \frac{2}{l} \int_0^l u_0 x (l-x) \sin\left(\frac{n\pi x}{l}\right) dx.$$

3. A tightly stretched string fixed at points, $x=0$ & $x=l$ is initially at a position given by $u(x) = u_0 \sin^2\left(\frac{\pi x}{l}\right)$. If it is released from x , find the displacement $u(x, t)$.

Ans. 1-D wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad 0 \leq x \leq l, \quad t \geq 0.$$

with b.c., $u(0, t) = u(l, t) = 0, \quad t \geq 0.$

g.i.c., $u(x, 0) = f(x) = u_0 \sin^2\left(\frac{\pi x}{l}\right) \quad \text{--- (1)}$

$$\frac{\partial u}{\partial t} \Big|_{t=0} = g(x) = 0.$$

$$u(x, t) = \sum_{n=1}^{\infty} c_n \cos\left(\frac{n\pi ct}{l}\right) \sin\left(\frac{n\pi x}{l}\right) \quad \text{--- (2)}$$

$$\text{where } c_n = \frac{2}{l} \int_0^l$$

$$\sin 3B = 3\sin B - 4\sin^3 B$$

Put $t=0$

$$(2) \Rightarrow u(x, 0) = \sum_{n=1}^{\infty} c_n \cdot 1 \cdot \sin\left(\frac{n\pi x}{l}\right) \quad \text{--- (3)}$$

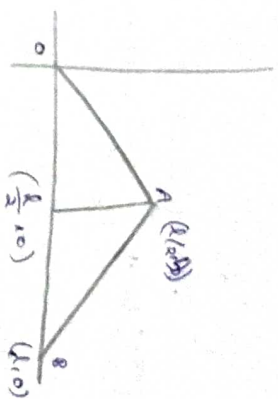
$$(1) \Rightarrow 2 \Rightarrow u_0 \sin^2\left(\frac{\pi x}{l}\right) = \sum_{n=1}^{\infty} c_n \sin\left(\frac{n\pi x}{l}\right)$$

$$\Rightarrow \frac{u_0}{4} \left[3 \sin\left(\frac{\pi x}{l}\right) - \sin\left(\frac{3\pi x}{l}\right) \right]$$

$$= c_1 \sin \frac{\pi x}{l} + c_2 \sin \frac{2\pi x}{l} + c_3 \sin \frac{3\pi x}{l} + c_4 \sin \frac{4\pi x}{l} + \dots$$

4. A string of length l is fastened at both ends. The mid point of the string is taken to a height h , then released from rest in that position. Find the displacement function $y(x)$.

Ans.



$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$$

B.C. are $y(0, t) = y(l, t) = 0.$

To find initial displacement / deflection of the string, we compute the equations of straight line

$$(x_1, y_1), (x_2, y_2).$$

Equation of AB

$$\frac{y-y_1}{y_2-y_1} = \frac{x-x_1}{x_2-x_1}$$

$$\frac{y-0}{h-0} = \frac{x-0}{l/2-0} \Rightarrow y = \frac{2h}{l} x, \quad 0 \leq x \leq l/2.$$

Eqn of AB

$$\frac{y-0}{h-0} = \frac{x-l}{l/2-l} \Rightarrow \frac{y}{h} = \frac{-2}{l} (x-l) \quad \text{--- (ii)}$$

$$\therefore y = \frac{-2h}{l} (x-l), \quad \frac{l}{2} \leq x \leq l.$$

$$f(x) = \frac{2hx}{l}, \quad 0 \leq x \leq l/2$$

$$-\frac{2h}{l} (x-l), \quad \frac{l}{2} \leq x \leq l.$$

$$g(x) = 0 \Rightarrow D_n = 0.$$

The solution is:

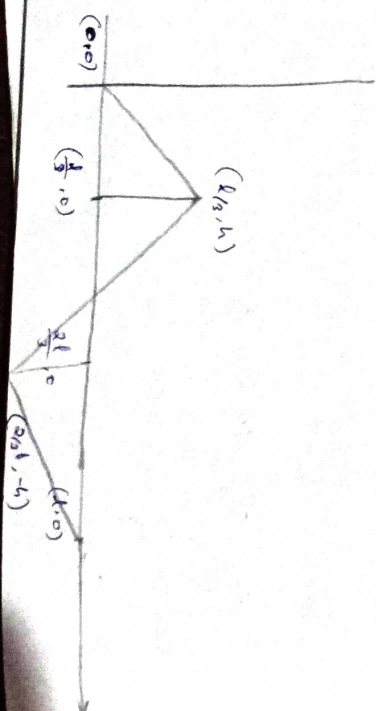
$$u(x, t) = \sum_{n=1}^{\infty} c_n \cos\left(\frac{n\pi x}{l}\right) \sin\left(\frac{n\pi x}{l}\right) \quad \text{--- (5)}$$

$$c_n = \frac{2}{l} \int_0^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{2}{l} \int_0^{l/2} \frac{2hx}{l} \sin\left(\frac{n\pi x}{l}\right) dx + \int_{l/2}^l \frac{-2h}{l} (x-l) \sin\left(\frac{n\pi x}{l}\right) dx$$

$$c_n = \frac{8h}{n^2 \pi^2 l} \sin\left(\frac{n\pi}{2}\right)$$

5 The points of trisection of a string are pulled aside through the same distance on opposite sides of the position of equilibrium & the string is released from rest. Derive an expression for the displacement at any instant t .



D'Alembert's solution of wave eqn.

If $f(x)$ & $g(x)$ are respectively the initial deflection & initial velocity of vibrating string then, 1-D wave equation is:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \quad \text{--- (1)}$$

The D'Alembert's solution is given by:

$$u(x, t) = \frac{1}{2} [f(x+ct) + f(x-ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds$$

Use D'Alembert's method to find displacement of a vibrating string with initial displacement $f(x) = \sin x$, $g(x) = 0$

$$\therefore u(x, t) = \frac{1}{2} [\sin(x+ct) + \sin(x-ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} 0 ds$$

$$= \frac{1}{2} [\sin(x+ct) + \sin(x-ct)]$$

$$u(x, t) = \frac{1}{2} [\sin(x+ct) + \sin(x-ct)] + \frac{0}{2c} [x+ct - x-ct]$$

$$= \frac{1}{2} [\sin(x+ct) + \sin(x-ct)]$$

1-D heat equation.

Consider a straight rod of constant cross-sectional area A and of homogeneous material placed along the x -axis from 0 to l . The temperature u of the rod depends on x & t . This phenomenon is described by 1-D heat equation.

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad 0 \leq x \leq l, \quad t \geq 0$$

where, $c^2 = \frac{k}{\rho c}$, Called the diffusivity of the material

where $k \Rightarrow$ thermal conductivity of material

$C \Rightarrow$ specific heat capacity of the material.

$\rho \Rightarrow$ density of the material.

Case-1: Ends of the rod kept at 0 temperature;

The solution of the heat equation

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad 0 \leq x \leq l, \quad t \geq 0.$$

subjected to BC,

$$u(0, t) = u(l, t) = 0, \quad t \geq 0$$

$$\& \text{ IC } \Rightarrow u(x, 0) = f(x), \quad 0 \leq x \leq l;$$

where $f(x)$ is the initial temperature distribution.

$$u(x, t) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{l}\right) e^{-\frac{c^2 n^2 \pi^2 t}{l^2}}$$

$$\text{where, } B_n = \frac{2}{l} \int_0^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx.$$

Case-2: Ends of the rod are kept insulated;

kept

When the ends of the rod are insulated, there is no heat flow across the ends with initial temp: $f(x)$ then, the boundary value of problem modelling the temperature distribution $u(x, t)$ is given as:

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad 0 \leq x \leq l, \quad t \geq 0$$

with boundary conditions

$$\left. \frac{\partial u}{\partial x} \right|_{x=0} = \left. \frac{\partial u}{\partial x} \right|_{x=l} = 0.$$

And IC: $u(x, 0) = f(x)$

Now the solution is given by;

$$u(x, t) = \sum_{n=0}^{\infty} A_n \cos\left(\frac{n\pi x}{l}\right) e^{-\frac{c^2 n^2 \pi^2 t}{l^2}}$$

$$\text{where, } A_0 = \frac{1}{l} \int_0^l f(x) dx$$

$$A_n = \frac{2}{l} \int_0^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

1. A homogeneous laterally insulated bar of length 100cm has its ends kept at zero temperature. Find the temperature distribution $u(x, t)$ if the initial temperature is $f(x)$

$$f(x) = \begin{cases} x & , 0 \leq x \leq 50 \\ 100-x & , 50 \leq x \leq 100. \end{cases}$$

Ans. let $u(x, t)$ be the temperature distribution. Then the 1-D

heat equation $\Rightarrow$

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad 0 \leq x \leq 100, \quad t \geq 0.$$

$$\text{BC } \Rightarrow u(0, t) = u(100, t) = 0.$$

$$\& \text{ IC } \Rightarrow f(x) \Rightarrow u(x, 0) = f(x) = \begin{cases} x & , 0 \leq x \leq 50 \\ 100-x & , 50 \leq x \leq 100 \end{cases}$$

The general solution is given as

$$u(x, t) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{100}\right) e^{-\frac{c^2 n^2 \pi^2 t}{100^2}}$$

$$\text{where, } B_n = \frac{2}{100} \int_0^{100} f(x) \sin\left(\frac{n\pi x}{100}\right) dx$$

$$= \frac{2}{100} \left[\int_0^{50} x \sin\left(\frac{n\pi x}{100}\right) dx + \int_{50}^{100} (100-x) \sin\left(\frac{n\pi x}{100}\right) dx \right]$$

$$= \frac{2}{100} \left[\left(\frac{x \cos \frac{n\pi x}{100}}{\frac{n\pi}{100}} + \frac{\sin \frac{n\pi x}{100}}{\frac{n\pi}{100}} \right) \Big|_0^{50} + \left(\frac{x \cos \frac{n\pi x}{100}}{\frac{n\pi}{100}} + \frac{\sin \frac{n\pi x}{100}}{\frac{n\pi}{100}} \right) \Big|_{50}^{100} \right]$$

$$+ \left[\frac{-100}{n\pi} (100-x) \cos \frac{n\pi x}{100} + \left(\frac{100}{n\pi} \right)^2 \sin \left(\frac{n\pi x}{100} \right) \right]_{50}^{100}$$

$$= \frac{1}{60} \int \frac{-100}{n\pi} x \cdot \cos \left(\frac{n\pi x}{100} \right) + \left(\frac{100}{n\pi} \right)^2 \sin \left(\frac{n\pi x}{100} \right) \Big|_0^{50}$$

$$+ \left[\frac{-100}{n\pi} (100-x) \cos \frac{n\pi x}{100} + \left(\frac{100}{n\pi} \right)^2 \sin \left(\frac{n\pi x}{100} \right) \right]_{50}^{100}$$

$$= \frac{1}{50} \left[\frac{-500}{n\pi} \cos \left(\frac{n\pi}{2} \right) + \left(\frac{100}{n\pi} \right)^2 \sin \left(\frac{n\pi}{2} \right) \right]$$

$$+ \left[0 + \frac{500}{n\pi} \cos \frac{n\pi}{2} + \left(\frac{100}{n\pi} \right)^2 \cdot 0 + \left(\frac{100}{n\pi} \right)^2 \sin \left(\frac{n\pi}{2} \right) \right]$$

$$= \frac{1}{50} \times 2 \left(\frac{100}{n\pi} \right)^2 \sin \left(\frac{n\pi}{2} \right)$$

$$= \frac{400}{n^2 \pi^2} \sin \left(\frac{n\pi}{2} \right)$$

$$\therefore u(x,t) = \sum_{n=1}^{\infty} \frac{400}{n^2 \pi^2} \sin \left(\frac{n\pi}{2} \right) \cdot \sin \left(\frac{n\pi x}{100} \right) e^{-\frac{c^2 n^2 \pi^2 t}{100^2}}$$

2. Find the temperature distribution in a laterally insulated bar of length l with both the ends insulated ~~at initial~~ temp be $\sin \left(\frac{\pi x}{l} \right)$

Ans. Let $u(x,t)$ be the temperature distribution & can be modelled by,

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad 0 \leq x \leq l, \quad t \geq 0.$$

$$\text{with BC: } \frac{\partial u}{\partial x} \Big|_{x=0} = \frac{\partial u}{\partial x} \Big|_{x=l} = 0, \quad t > 0.$$

$$\text{IC: } u(x,0) = f(x) = \sin \left(\frac{\pi x}{l} \right)$$

$$u(x,t) = \sum_{n=0}^{\infty} A_n \cos \left(\frac{n\pi x}{l} \right) e^{-\frac{c^2 n^2 \pi^2 t}{l^2}}$$

$$A_0 = \frac{1}{l} \int_0^l \sin \left(\frac{\pi x}{l} \right) dx = \frac{1}{l} \left[\frac{-l}{\pi} \cos \left(\frac{\pi x}{l} \right) \right]_0^l$$

$$= \frac{-1}{\pi} [\cos \pi - \cos 0] = \frac{2}{\pi}$$

$$A_n = \frac{2}{l} \int_0^l \sin \left(\frac{\pi x}{l} \right) \cdot \cos \left(\frac{n\pi x}{l} \right) dx$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B.$$

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$\cos(A-B) = \cos A \cos B + \sin A \sin B$$

$$\cos(A-B) - \cos(A+B) = 2 \sin A \sin B.$$

$$2 \sin A \cdot \cos B = \sin(A+B) + \sin(A-B)$$

$$\therefore A_n = \frac{1}{l} \int_0^l 2 \sin \left(\frac{\pi x}{l} \right) \cos \left(\frac{n\pi x}{l} \right) dx.$$

$$= \frac{1}{l} \int_0^l \left[\sin \left(\frac{\pi x}{l} + \frac{n\pi x}{l} \right) + \sin \left(\frac{\pi x}{l} - \frac{n\pi x}{l} \right) \right] dx$$

$$= \frac{1}{l} \int_0^l \frac{-1}{(l+n)\pi} \cos \frac{(l+n)\pi x}{l} - \frac{1}{(l-n)\pi} \cos \frac{(l-n)\pi x}{l} \Big|_0^l$$

$$= \frac{1}{l} \frac{-1}{(l+n)\pi} \left[\cos \frac{(l+n)\pi x}{l} - 1 \right] - \frac{1}{(l-n)\pi} \left[\cos \frac{(l-n)\pi x}{l} - 1 \right]$$

$$= \frac{-1}{l(n)\pi} \cos(l+n)$$

$$= \frac{-1}{(l+n)\pi} [(-1)^{n+1} - 1] - \frac{1}{(l-n)\pi} [(-1)^{1-n} - 1]$$

$$= \frac{(-1)^{n+1} - 1}{\pi} \left[\frac{1}{l+n} - \frac{1}{l-n} \right]$$

Imp. 3. An insulated rod of length L has its ends A & B maintained at 0°C & 100°C respectively, until steady state condition prevails. If B is suddenly reduced to 0°C & maintained at 0°C then, find the temperature in the rod at distance x from A at time t .

Ans. Let $u(x, t)$ be the temperature distribution & can be modelled by

$$\frac{\partial u}{\partial t} = c^2 \cdot \frac{\partial^2 u}{\partial x^2} \quad 0 \leq x \leq L, \quad t \geq 0.$$

with BC $\Rightarrow u(0, t) = u(L, t) = 100^\circ\text{C}$ $t \geq 0$

IC $\Rightarrow u(x, 0) = f(x) = ?$ $0 \leq x \leq L$

In steady state, the temp distribution is independent of time. i.e.,

$$\frac{\partial u}{\partial t} = 0. \quad \therefore \frac{\partial^2 u}{\partial x^2} = 0$$

($\because u$ is independent of t)
K PDE $\approx$ ODE

$$\Rightarrow \frac{d^2 u}{dx^2} = 0.$$

On integrating w.r.t x ,

$$\frac{du}{dx} = a_1$$

integrating w.r.t x ,

$$u = a_2 x + b.$$

Given,

$$u(0, t) = 0^\circ\text{C}, \quad u(L, t) = 100^\circ\text{C}.$$

$$u(0, t) = 0 \Rightarrow b = 0$$

$$u(L, t) = 100 \Rightarrow aL + b = aL + 0 = 100.$$

$$\therefore a = \frac{100}{L}$$

$$\therefore u(x, t) = \frac{100}{L} x \Rightarrow f(x).$$

Thus the IC $\Rightarrow u(x, 0) = f(x) = \frac{100}{L} x$, $0 \leq x \leq L$
BC $\Rightarrow u(0, t) = u(L, t) = 0$.

Then the solution is given by;

$$u(x, t) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) e^{-\frac{c^2 n^2 \pi^2 t}{L^2}}; \quad 0 \leq x \leq L; \quad t \geq 0$$

where, $B_n = \frac{2}{L} \int_0^L \frac{100}{L} x \sin\left(\frac{n\pi x}{L}\right) dx.$

$$= \frac{200}{L^2} \int_0^L x \cdot \sin\left(\frac{n\pi x}{L}\right) dx$$

$$= \frac{200}{L^2} \left[\frac{xL}{n\pi} \cos \frac{n\pi x}{L} + \sin \frac{n\pi x}{L} \cdot \left(\frac{L}{n\pi}\right)^2 \right]_0^L$$

$$= \frac{200}{L^2} \left[\frac{L^2}{n\pi} \cos(n\pi) - \sin(n\pi) \cdot \frac{L^2}{n\pi^2} \right]$$

$$= [0]$$

$$= \frac{200}{L^2} \cdot \frac{L^2}{n\pi} \left[-\cos n\pi - \sin(n\pi) \cdot \frac{L}{n\pi} \right]$$

$$= \frac{200}{n\pi} \left[-\cos n\pi \right] \frac{\sin(n\pi)}{n\pi}$$

$$= \frac{200}{n\pi} (\cos n\pi)$$

$$= \frac{200}{n\pi} (-1)^{n+1}$$

Laplace Equation

Solution of Laplace equations in 2-D.

Laplace equation in 2-D is :

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \text{--- (1)}$$

By applying the method of separation of variables, we obtain various possible solutions of Laplace equations, & they are given by :

$$u(x, y) = (c_1 x + c_2) (c_3 y + c_4)$$

$$\text{or } u(x, y) = [c_1 e^{px} + c_2 e^{-px}] [c_3 e^{py} + c_4 e^{-py}]$$

$$\text{or } u(x, y) = (c_1 \cos px + c_2 \sin px) [c_3 e^{py} + c_4 e^{-py}]$$

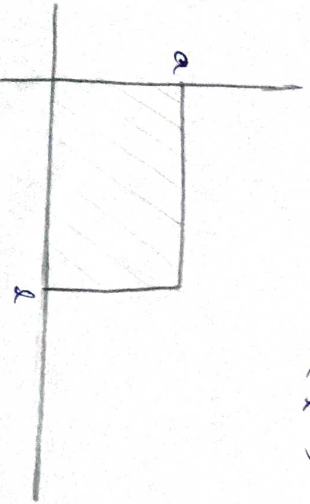
Of these 3 solutions, we have to choose that solution which is consistently the physical nature of the problem & gives boundary conditions.

1. Use separation of variables method to solve the equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$\text{subject to BC } \rightarrow u(0, y) = u(l, y) = u(x, 0) = 0.$$

$$u(x, a) = \sin\left(\frac{n\pi x}{l}\right)$$



The possible solutions are :

$$I \rightarrow u(x, y) = (c_1 x + c_2) (c_3 y + c_4)$$

$$II \rightarrow u(x, y) = [c_1 e^{px} + c_2 e^{-px}] [c_3 e^{py} + c_4 e^{-py}]$$

$$III \rightarrow u(x, y) = (c_1 \cos px + c_2 \sin px) [c_3 e^{py} + c_4 e^{-py}]$$

$$u(x, y) = 0 \quad (\text{Given})$$

Consider case I

$$u(x, y) = (c_1 x + c_2) (c_3 y + c_4)$$

$$\therefore u(0, y) = 0$$

$$\Rightarrow c_1(0) = 0$$

$$u(0, y) = 0$$

$$x(0) \cdot y(y) = 0$$

$$x(0) \cdot y(y) = 0$$

If y is zero, u will always be zero for any value of x, y .

$$x(0) = 0 \Rightarrow c_1 x + c_2 = 0$$

$$\Rightarrow c_2 = 0.$$

$$\text{Consider, } u(l, y) = 0$$

$$u(l, y) = 0.$$

$$x(l) = 0 \Rightarrow c_1 l + 0 = 0$$

$$x(l) \cdot y(y) = 0.$$

$$\text{if } y(y) = 0 \Rightarrow u = 0$$

$$\therefore x(x) = 0.$$

$$\therefore \text{not possible}$$

$$\text{so } x(l) = 0.$$

$\rightarrow$ Consider case II :

$$x(0) = 0 \Rightarrow c_1 + c_2 = 0 \Rightarrow c_1 = -c_2.$$

$$x(l) = 0 \Rightarrow c_1 e^{pl} + c_2 e^{-pl} = 0$$

$$c_1 e^{pl} - c_1 e^{-pl} = 0$$

$$c_1 [e^{pl} - e^{-pl}] = 0$$

$$\Rightarrow c_1 = 0 \Rightarrow c_2 = 0.$$

$$\therefore x = 0.$$

→ Consider case III :

$$X(x) = C_1 \cosh px + C_2 \sinh px$$

$$X(0) = 0 \Rightarrow C_1 \cdot 1 + C_2 \cdot 0 = 0.$$

$$\therefore C_1 = 0.$$

$$X(x) = C_2 \sinh px$$

$$X(l) = 0 \Rightarrow C_2 \sinh pl = 0.$$

$$\Rightarrow \sinh pl = 0$$

$$\sinh pl = 0.$$

$$\Rightarrow pl = n\pi \Rightarrow p = \frac{n\pi}{l}$$

$$Y = C_3 e^{py} + C_4 e^{-py}.$$

$$Y(0) = 0$$

$$C_3 e^0 + C_4 e^0 = 0$$

$$C_3 = -C_4 \text{ or } C_4 = -C_3$$

$$Y = C_3 \left[e^{py} - e^{-py} \right]$$

$$= C_3 \left[e^{\frac{n\pi}{l}y} - e^{-\frac{n\pi}{l}y} \right]$$

$$X = C_3 \sinh \frac{n\pi}{l} x$$

$$Y = C_3 \left[e^{\frac{n\pi}{l}x} - e^{-\frac{n\pi}{l}x} \right]$$

$$u(x,y) = XY$$

$$= C_3 C_3 \sinh \left(\frac{n\pi}{l} x \right) \left[e^{\frac{n\pi}{l}y} - e^{-\frac{n\pi}{l}y} \right]$$

$$= C_6 \sinh \left(\frac{n\pi}{l} x \right) \left[e^{\frac{n\pi}{l}y} - e^{-\frac{n\pi}{l}y} \right] \quad \text{--- (1).}$$

$$u(x,a) = \sinh \left(\frac{n\pi x}{l} \right) \quad \text{--- (2).}$$

$$C_n \sinh \left(\frac{n\pi x}{l} \right) \left[e^{\frac{n\pi y}{l}} - e^{-\frac{n\pi y}{l}} \right] = \sinh \frac{n\pi x}{l}$$

$$\Rightarrow C_n \Rightarrow \frac{1}{\frac{e^{\frac{n\pi a}{l}} - e^{-\frac{n\pi a}{l}}}{2 \sinh \left(\frac{n\pi a}{l} \right)}} = \frac{1}{2 \sinh \left(\frac{n\pi a}{l} \right)}$$

Thus solution is,

$$u(x,y) = \frac{1}{2 \sinh \left(\frac{n\pi a}{l} \right)} \sinh \left(\frac{n\pi x}{l} \right) \left[e^{\frac{n\pi y}{l}} - e^{-\frac{n\pi y}{l}} \right]$$

$$= \frac{\sinh \left(\frac{n\pi y}{l} \right)}{\sinh \left(\frac{n\pi a}{l} \right)} \sinh \left(\frac{n\pi x}{l} \right)$$

$$\underline{\underline{\sinh \left(\frac{n\pi y}{l} \right) \sinh \left(\frac{n\pi x}{l} \right)}}$$